

Breast Cancer Detection using Machine Learning

A Comparative Analysis of Decision Trees and SVM

Machine Learning Project Group

January 28, 2026

Abstract. This report presents the design, implementation, and evaluation of a machine learning system for the diagnosis of breast cancer based on the Breast Cancer Wisconsin (Diagnostic) dataset. The primary objective is to develop a decision support system capable of classifying tumor masses as Benign (0) or Malignant (1) with high accuracy. Particular emphasis is placed on minimizing False Negatives (FN), as missing a malignant case typically carries the most severe consequences in a clinical setting. We compare the performance of an interpretable model (Decision Tree) against a high-dimensional classifier (Support Vector Machine), utilizing Principal Component Analysis (PCA) for dimensionality reduction. Our experimental results demonstrate that while SVM achieves superior quantitative metrics, the Decision Tree offers valuable qualitative insights into the critical geometric thresholds of cell nuclei.

Contents

1	Description of the Domain and Objectives	3
1.1	Objectives	3
2	Exploratory Data Analysis (EDA)	3
2.1	Dataset Structure and Preview	3
2.2	Encoding and Class Distribution	3
2.3	Multidimensional Feature Analysis	4
2.4	Multicollinearity Analysis	6
3	Feature Engineering: Principal Component Analysis	7
3.1	Data Standardization	7
3.2	Dimensionality Reduction (PCA)	8
3.2.1	Variance Retention Strategy (95% Threshold)	8
3.2.2	Cumulative Explained Variance	8
3.3	Feature Space Transformation	9
3.4	Visualization of the Latent Space	9
3.5	Methodology for Data Leakage Prevention	10

4	Experimental Results: Decision Tree	11
4.1	Validation Strategy: Addressing Data Scarcity	11
4.2	Regularization via Pre-Pruning (Maximum Depth)	11
4.3	K-Fold Cross-Validation Analysis	12
4.3.1	Tree Stability Across Folds	12
4.3.2	K-Fold Performance Metrics	13
4.4	Bias-Variance Analysis	14
4.5	Leave-One-Out (LOO) Analysis	14
4.5.1	LOO Performance Metrics	14
4.6	Post-Pruning Optimization (Cost-Complexity)	15
4.6.1	Searching for the Optimal Alpha	15
4.6.2	Optimal Model Selection	16
5	Experimental Results: Support Vector Machine	17
5.1	The Kernel Trick	17
5.2	Input Feature Space	18
5.3	Kernel Analysis and Cross-Validation	18
5.4	Final Test Prediction and Comparison	19
5.5	ROC Curve Analysis	19
6	Comparative Analysis: Decision Tree vs SVM	20
6.1	Consolidated Final Metrics	20
6.2	Statistical Credibility and Significance	21
6.3	Model Robustness and Generalizability	21
7	Conclusions	21

1. Description of the Domain and Objectives

The domain of this project is medical diagnostics, specifically within the field of oncology. Breast cancer remains one of the most common cancers worldwide, and early detection is crucial for improving survival rates.

1.1. Objectives

The primary goal is to automate the classification of breast masses based on biometric data derived from digitized images of a Fine Needle Aspirate (FNA). The classification task is binary:

- **Benign (0)**: Non-cancerous cells.
- **Malignant (1)**: Cancerous cells requiring immediate medical intervention.

From a clinical perspective, not all classification errors have equal weight. A **False Negative** (classifying a malignant tumor as benign) is a critical error that could delay life-saving treatment. Therefore, our technical objective is to maximize **Recall (Sensitivity)** for the Malignant class while maintaining a high overall **Accuracy**.

2. Exploratory Data Analysis (EDA)

Before applying any learning algorithm, a thorough Exploratory Data Analysis was conducted to understand the underlying structure of the data, the separation of classes, and the relationships between features.

2.1. Dataset Structure and Preview

The dataset utilized is the **Breast Cancer Wisconsin (Diagnostic) Data Set**. It consists of 569 instances, each characterized by 30 numeric features representing the Mean, Standard Error, and “Worst” (mean of the three largest values) of 10 cell nuclei characteristics (e.g., **radius**, **texture**, **area**).

Table 1 provides a snapshot of the raw data, highlighting the difference in scale between geometric features (like **Area**, reaching > 1000) and morphological coefficients (like **Smoothness**, < 0.1), which necessitates subsequent standardization.

Table 1: Dataset Preview: First 5 rows (Selected columns)

Index	radius1	texture1	perimeter1	area1	smoothness1	...	radius3	texture3	perimeter3	area3
0	17.99	10.38	122.80	1001.0	0.11840	...	25.38	17.33	184.60	2019.0
1	20.57	17.77	132.90	1326.0	0.08474	...	24.99	23.41	158.80	1956.0
2	19.69	21.25	130.00	1203.0	0.10960	...	23.57	25.53	152.50	1709.0
3	11.42	20.38	77.58	386.1	0.14250	...	14.91	26.50	98.87	567.7
4	20.29	14.34	135.10	1297.0	0.10030	...	22.54	16.67	152.20	1575.0

2.2. Encoding and Class Distribution

To integrate the diagnosis into the mathematical models, a **Target Encoding** step was performed before the train-test split: Benign samples were mapped to **0** and Malignant samples to **1**.

An analysis of the target variable reveals a class distribution of **357 Benign (62.7%)** and **212 Malignant (37.3%)** samples on the entire dataset.

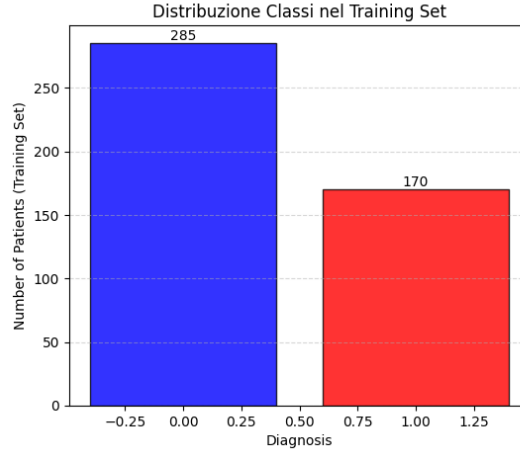


Figure 1: Distribution of Diagnosis: Benign vs Malignant on Training Set

From a theoretical perspective, this indicates a mild class imbalance. While not severe enough to require synthetic oversampling (like SMOTE), it introduces a potential **Sampling Bias** if not addressed during the evaluation phase. If a naive classifier were to predict all cases as Benign, it would achieve an accuracy of $\approx 63\%$. To mitigate this, we employed **Stratified Cross-Validation**, which ensures each training fold preserves the 63/37 ratio, providing a more credible estimation of the model's true generalization performance on the minority class.

2.3. Multidimensional Feature Analysis

To identify the biometric markers with the highest diagnostic potential, we performed a combined analysis of feature distributions, scales, and separability.

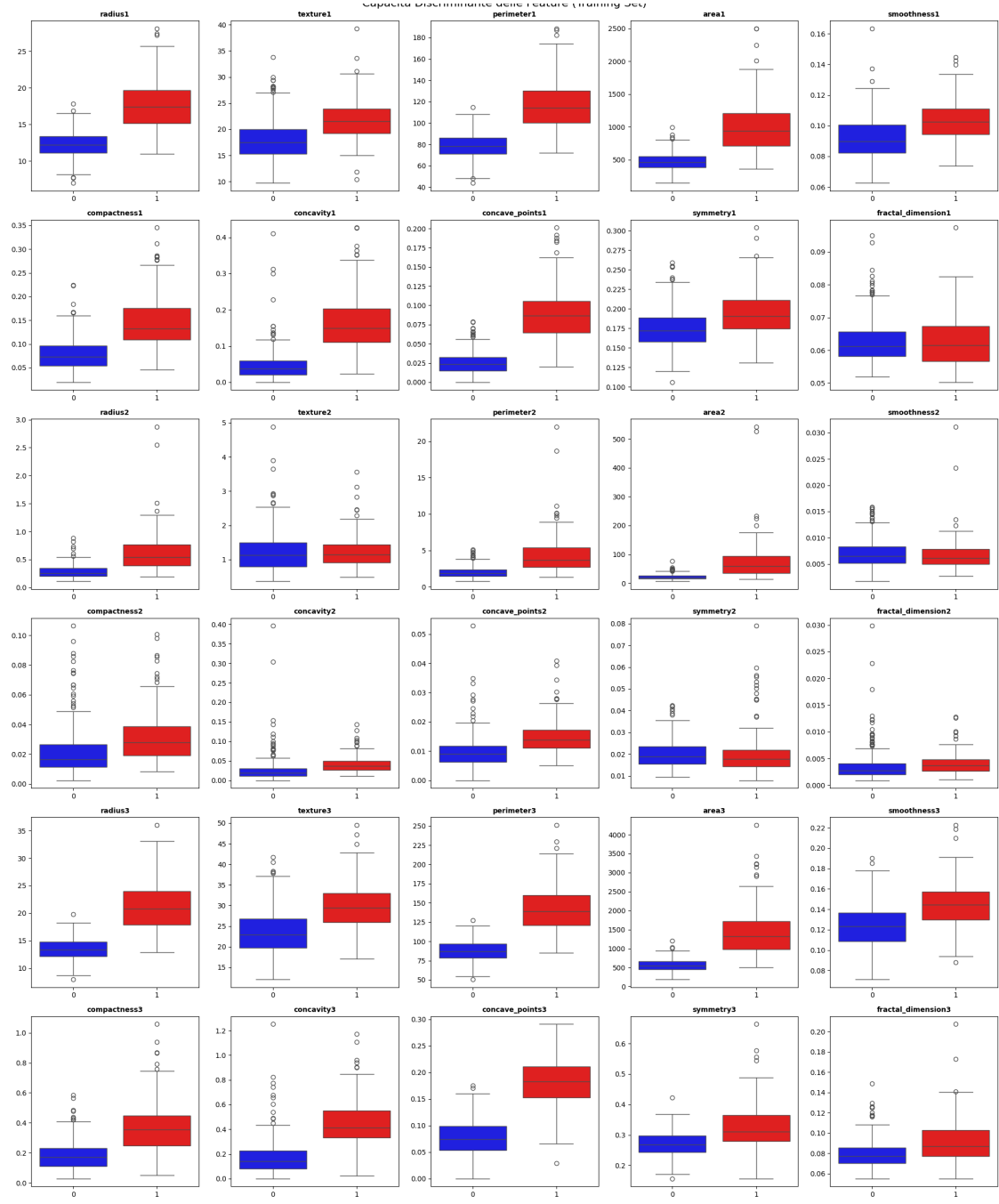


Figure 2: Comprehensive Boxplot Analysis. The plot reveals (1) extreme scale disparity across features, (2) the presence of rare morphological outliers (dots), and (3) the discriminative power of features like Radius and Concavity for benign/malignant separation.

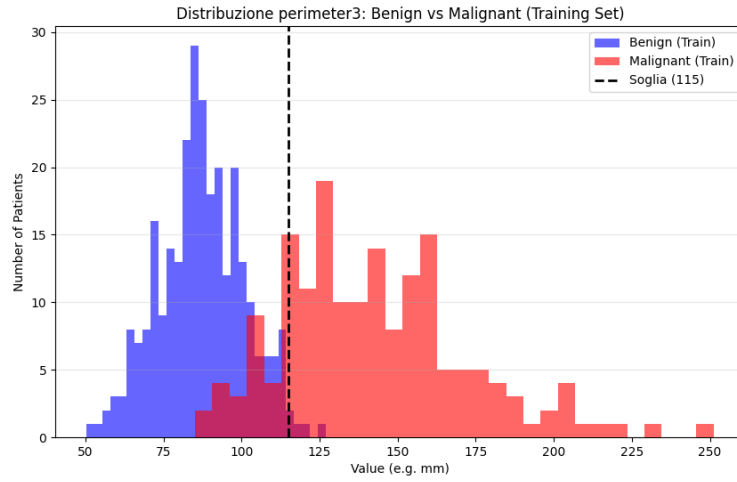


Figure 3: Scatter plot visualizing the distribution of Benign vs Malignant cases.

The analysis of Figure 2 reveals several critical insights:

1. **Scale Imbalance and Preprocessing Requirements:** Physical dimensions (e.g., `area_mean`, $\mu \approx 654$) operate at three orders of magnitude higher than coefficients like `smoothness_mean` ($\mu \approx 0.096$). This magnitude gap confirms the necessity of Z-score standardization ($\mu = 0, \sigma = 1$) to prevent biased gradient updates in distance-based models.
2. **Outlier Presence and Robustness:** Using a threshold of $|z| > 3$, significant outliers were detected in features like `compactness_se`. These extreme values represent rare tumor variants. Their presence justifies the use of the **RBF Kernel**, which uses a localized radial influence that is less susceptible to being skewed by distant outliers compared to a linear hyperplane.
3. **Feature Discriminative Power (Separability):**
 - *High Separability:* Features such as `concave points_mean` and `radius_worst` show bi-modal distributions with minimal overlap between classes.
 - *Low Separability:* Biological noise is higher in features like `fractal_dimension` or `texture_se`, where the distributions for B and M classes almost completely overlap.

2.4. Multicollinearity Analysis

Given the geometric origin of the features (e.g., Radius is mathematically related to Perimeter and Area), we expected a high degree of correlation. We computed the Pearson Correlation Matrix to verify this.

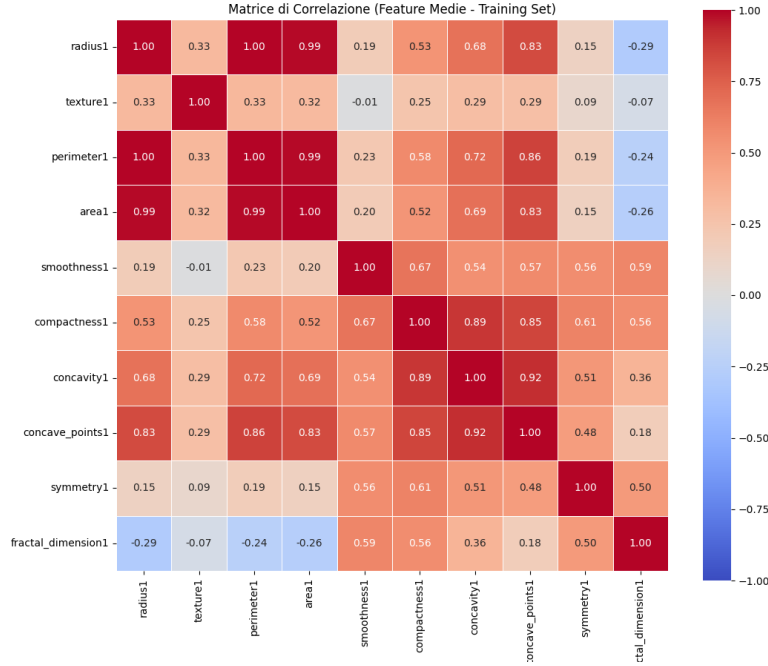


Figure 4: Correlation Heatmap of the features. Darker red indicates high positive correlation.

Figure 4 confirms the presence of **Multicollinearity**, a state where independent variables are highly correlated.

1. **Structural Redundancy:** We observed near-perfect correlation ($\rho \approx 1.0$) between **radius**, **perimeter**, and **area**, which is expected given their geometric interdependence.
2. **Statistical Implications:** Multicollinearity increases the variance of coefficient estimates, making models like Logistic Regression or unpruned Decision Trees unstable.

This redundant structure provides the theoretical justification for **Principal Component Analysis (PCA)**. By performing an eigenvalue decomposition of the covariance matrix, we can project the 30-dimensional space into a lower-dimension manifold where each component is **orthogonal** (uncorrelated). This transformation is essential for the SVM algorithm, as it reduces the complexity of finding the optimal hyperplane without losing the structural information of the data.

3. Feature Engineering: Principal Component Analysis

Feature engineering is a critical step in this pipeline, aimed at transforming the 30 highly correlated raw features into a reduced set of uncorrelated variables that capture the maximum information (variance).

3.1. Data Standardization

PCA calculates the maximum directions of variance. Since variance is sensitive to the scale of the variable (e.g., **Area** ≈ 1000 vs **Smoothness** ≈ 0.1), applying PCA on raw data would bias the components heavily towards features with larger magnitudes.

To address this, we applied **Z-score Standardization** using `StandardScaler`:

$$z = \frac{x - \mu}{\sigma}$$

Crucially, to prevent **data leakage**, the scaler was fitted **only on the Training Set**. The parameters μ (mean) and σ (standard deviation) learned from the training data were then applied to transform the Test Set.

3.2. Dimensionality Reduction (PCA)

We performed PCA to identify the eigenvectors (Principal Components) of the covariance matrix.

3.2.1 Variance Retention Strategy (95% Threshold)

Given the critical nature of medical diagnostics, maximizing information retention is paramount. The Kaiser Rule ($\lambda > 1$) suggested retaining only 6 components. However, this would discard approximately 11% of the variance, potentially containing subtle but clinically relevant signal.

Therefore, we adopted a more conservative **Variance Retention Strategy**, selecting the number of components required to explain at least **95%** of the total variance. This threshold resulted in the selection of **10 Principal Components**. This choice ensures that the vast majority of the biological information is preserved while still achieving a 3x dimensionality reduction (from 30 to 10).

3.2.2 Cumulative Explained Variance

We verified this choice by analyzing the cumulative variance explained ratio.

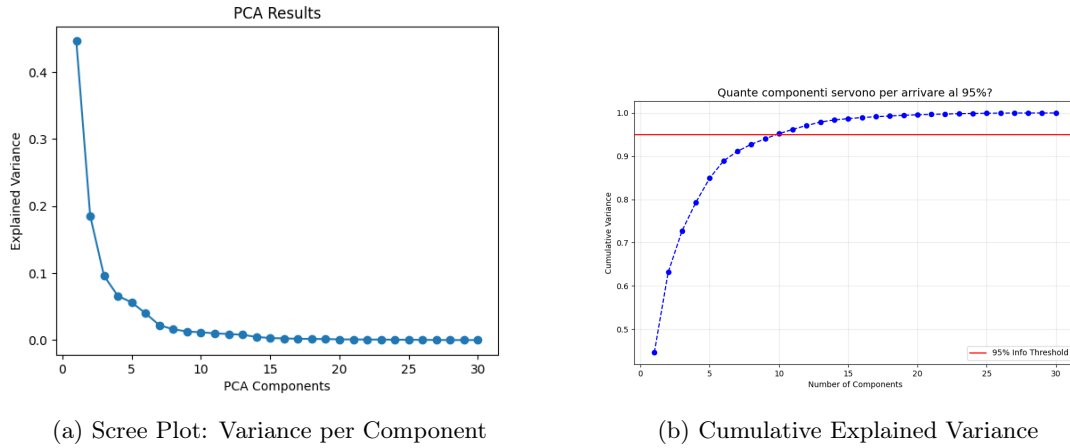


Figure 5: Analysis of Explained Variance. The Scree plot shows the "elbow" indicating the optimal number of components.

The first 10 components explain approximately **95.21%** of the total information contained in the dataset. This represents a significant increase in information retention compared to the 89% of the first 6 components, providing a richer feature set for the classification models.

3.3. Feature Space Transformation

The final transformation projected the dataset onto the 10-dimensional subspace defined by the top 10 eigenvectors.

- **Training Data:** Transformed from 30 dimensions to 10.
- **Test Data:** Scaled and projected using the training set's eigenvectors.

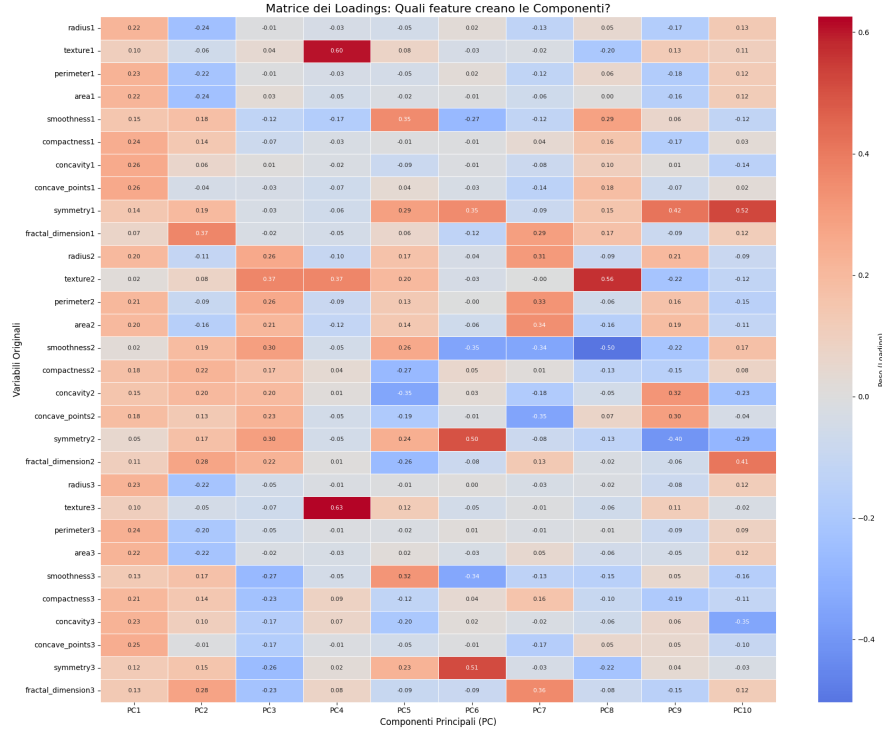


Figure 6: PCA Loadings Heatmap. This visualization reveals which original features contribute most to each Principal Component.

The following table shows the first 5 rows of the transformed dataset, illustrating the values in the new 10-dimensional feature space:

Table 2: First 5 rows of the PCA-transformed Feature Matrix

Index	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
0	-0.876	-2.571	-0.549	1.636	-0.263	0.110	0.506	-0.214	-0.109	0.417
1	-2.505	0.192	-0.405	-2.448	0.601	0.008	-1.094	0.073	0.555	0.156
2	-1.354	0.418	2.545	1.219	-1.767	0.709	0.767	-0.053	-0.437	0.186
3	4.854	3.018	-1.627	0.702	-2.814	-0.375	-0.474	0.061	0.180	-0.564
4	2.926	-1.867	2.486	-0.104	0.181	-0.975	-0.773	-0.177	-0.220	-0.343

3.4. Visualization of the Latent Space

To assess the discriminative power of the new features, we visualized the dataset in the 2D space defined by the first two components (PC1 and PC2), which alone account for over 64% of the variance.

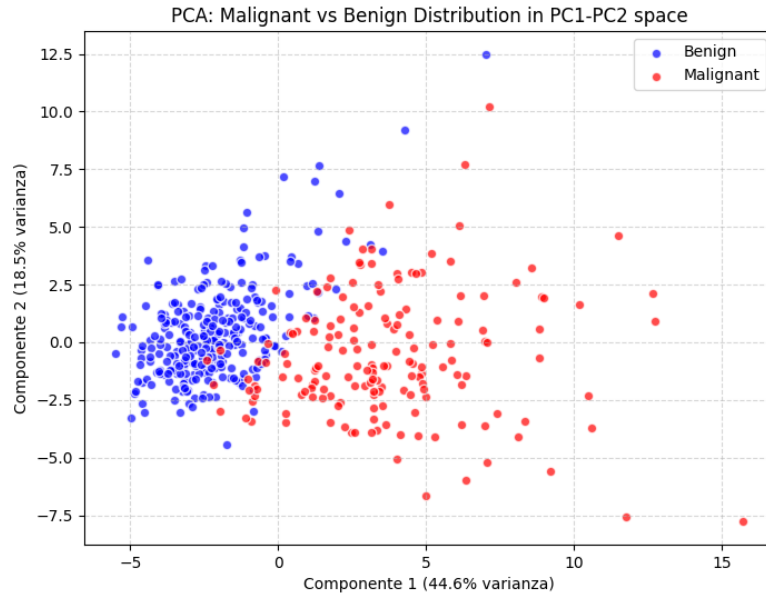


Figure 7: Scatter plot of Malignant (Red) vs Benign (Blue) cases in the PC1-PC2 space. The clear but imperfect separation justifies the use of non-linear models.

As shown in Figure 7, the two classes exhibit a reasonably clear separation even in just two dimensions. The Malignant cases tend to cluster towards the right (positive PC1), while Benign cases cluster to the left. However, a region of overlap persists, suggesting that a simple linear classifier might be insufficient and motivating the use of non-linear models like the RBF SVM.

3.5. Methodology for Data Leakage Prevention

A critical challenge in developing clinical diagnostic models is ensuring the rigorous separation between training knowledge and test evaluation. **Data Leakage** occurs when information from outside the training dataset is used to create the model, leading to overly optimistic results that fail in real-world deployment.

In this project, we implemented a strict “**Fit on Train, Transform on Test**” protocol to maintain preprocessing integrity:

1. **Standardization Integrity:** The `StandardScaler` was fitted exclusively on the training set. The resulting means (μ) and standard deviations (σ) were then applied to the test set as a fixed transformation. This prevents the training process from “seeing” the distribution or outliers of the test population.
2. **PCA Projection Integrity:** Similarly, the Principal Components (eigenvectors) were derived solely from the training covariance matrix. The test data was projected onto this established latent space without re-calculating the components. This ensures that the model evaluates the test samples within the exact same coordinate system defined during training.
3. **Process State Management:** During the optimization phase, we identified a “Process Leakage” risk where conditional guards (e.g., `if X_test_pca not in locals()`) could lead to the use of stale or misaligned features. We resolved this by enforcing an **Explicit**

Procedural Pipeline, ensuring that every test evaluation is preceded by a mandatory transformation using the current training state.

These measures ensure that our performance metrics (Accuracy, Recall) are scientifically valid and represent a reliable estimate of the system’s performance on future, unseen diagnostic cases.

4. Experimental Results: Decision Tree

The Decision Tree was selected as our baseline model due to its “White-Box” nature, allowing for full interpretability of the decision logic.

4.1. Validation Strategy: Addressing Data Scarcity

The dataset consists of 569 samples. In the context of robust machine learning validation, this is considered a **small dataset**. While a traditional hold-out method (e.g., a fixed 70/30 train-test split) might yield apparently high accuracy, it is statistically fragile in this context due to high variance.

To mitigate this statistical unreliability and ensure the robustness of our diagnostic system, we adopted rigorous resampling methods: **Stratified 10-Fold Cross-Validation** and **Leave-One-Out (LOO) Validation**.

4.2. Regularization via Pre-Pruning (Maximum Depth)

To manage the bias-variance trade-off during the initial construction of the tree, we implemented a **Pre-pruning** (Early Stopping) strategy by setting `max_depth=3`.

In the context of the “Evaluation and Credibility” framework, this choice is justified by:

1. **High Variance Prevention:** Given our small dataset ($N = 569$), allowing the tree to grow to its full potential (until leaves are pure) would lead to extreme overfitting. A depth of 3 constrains the model to capture only the most significant biometric thresholds (e.g., `perimeter_worst`, `concave_points_mean`) while ignoring idiosyncratic noise in the training samples.
2. **Clinical Interpretability:** A diagnostic tool must be auditable. A shallow tree ensures that the decision logic remains transparent and easy for a clinician to verify, acting as a “white-box” model.
3. **Generalization Stability:** Restricting depth ensures that the decision boundaries are orthogonal and stable across different folds, as verified in Figure 9.

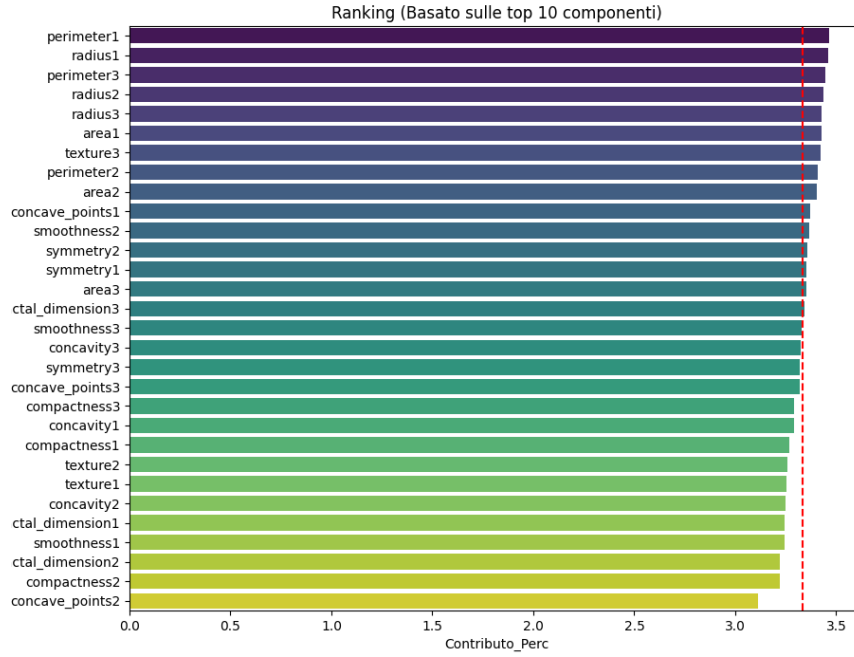


Figure 8: Feature Importance Ranking extracted from the Decision Tree. It confirms that geometric properties like 'Worst Perimeter' are the most discriminative.

4.3. K-Fold Cross-Validation Analysis

We executed 10 iterations of the training process. In each iteration, the model (max depth = 3) was trained on 9 folds and validated on the remaining one.

4.3.1 Tree Stability Across Folds

To verify the stability of the decision boundaries, we visualized the resulting tree structure for each of the 10 folds.

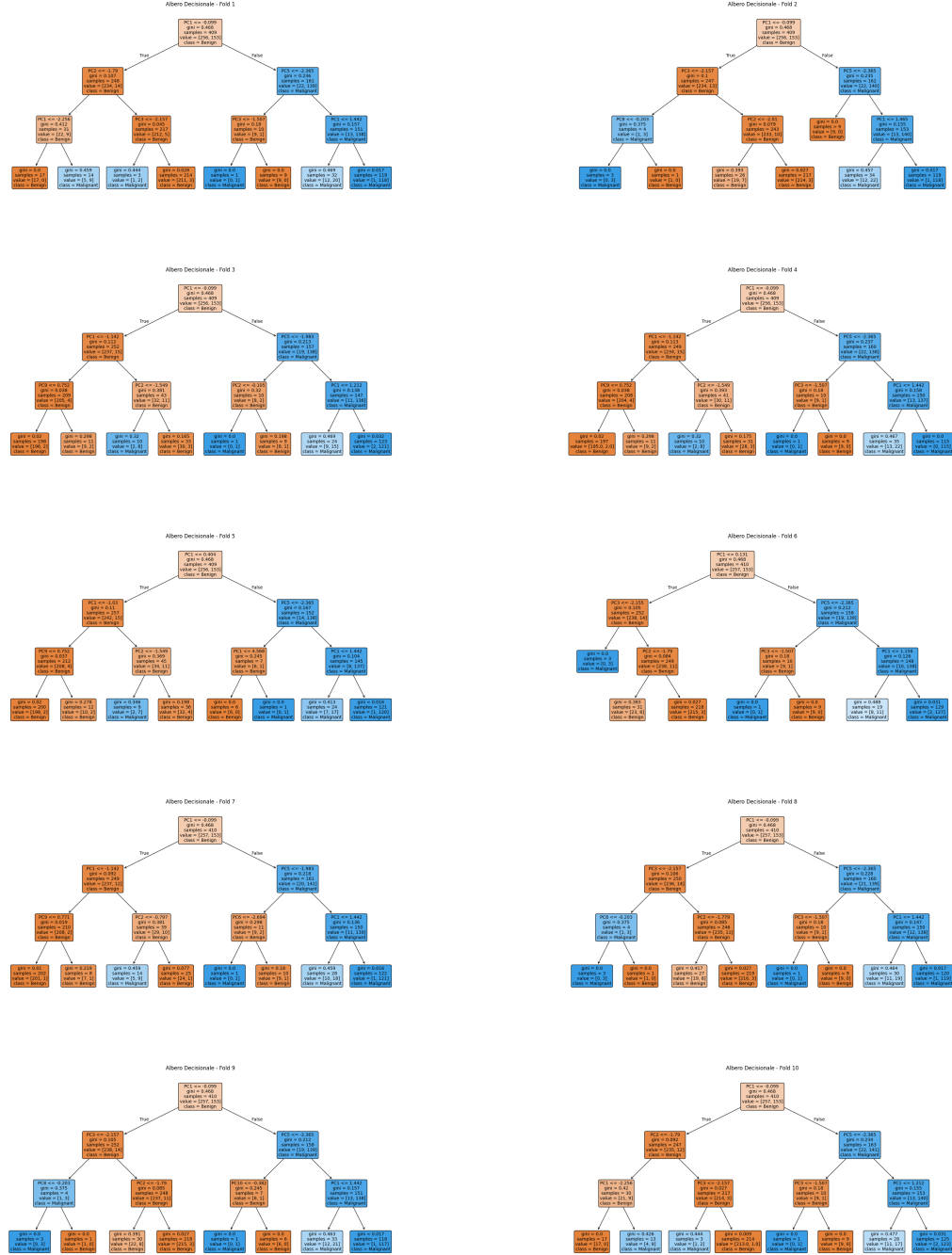


Figure 9: Decision trees generated across the 10 folds. The consistency in feature selection (splitting on similar biometric markers) demonstrates the reliability of our model despite the limited dataset.

4.3.2 K-Fold Performance Metrics

The aggregated performance metrics across the 10 folds indicate a robust baseline model. The 95% Confidence Interval (CI) suggests that we can expect the model's accuracy on unseen data to fall within this range with high probability.

Table 3: Stratified 10-Fold Cross-Validation Performance

Metric	Value
Mean Accuracy	91.85%
95% Confidence Interval	[89.04%, 94.65%]
Mean F1-Score	89.13%

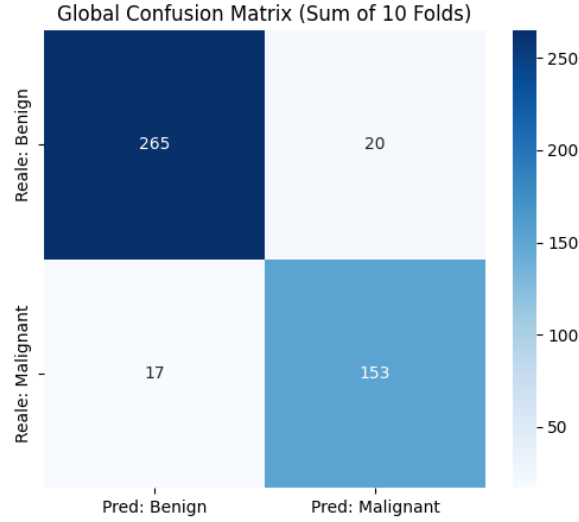


Figure 10: Global Confusion Matrix: Aggregated results of the 10 prediction rounds (Decision Tree). This matrix shows the total TP, TN, FP, and FN across all folds.

4.4. Bias-Variance Analysis

An analysis of the model performance dynamics reveals the relationship between training size and model error:

- **High Variance Check:** The gap between training and validation scores for both models narrows significantly as training examples increase, indicating that the regularization (Pruning for Tree, C parameter for SVM) is effectively preventing severe overfitting.
- **Sample Size Sufficiency:** The plateau in performance suggests that adding more medical instances would yield diminishing returns, confirming the 569 instances are adequate for this feature space.

4.5. Leave-One-Out (LOO) Analysis

Following the K-Fold analysis, we performed a Leave-One-Out validation to maximize the training set size ($N - 1$ samples) for each prediction, reducing the bias of the performance estimate.

4.5.1 LOO Performance Metrics

The results obtained from the LOO validation confirm the findings of the K-Fold strategy, showing consistent reliability metrics.

Table 4: Leave-One-Out (LOO) Validation Performance

Metric	Value
Mean Accuracy	89.67%
95% Confidence Interval	[86.86%, 92.48%]
Mean F1-Score	86.38%

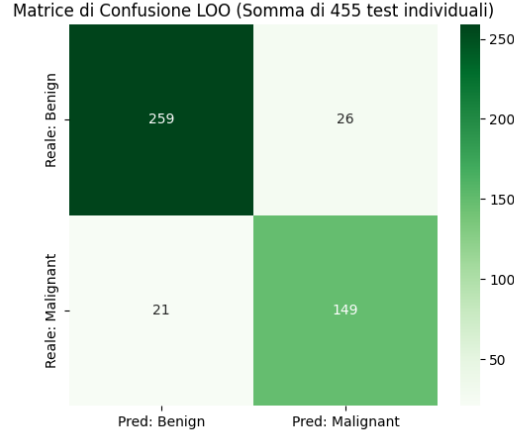


Figure 11: Leave-One-Out Confusion Matrix: Result of N individual tests.

4.6. Post-Pruning Optimization (Cost-Complexity)

Decision trees, by their nature, tend to grow indefinitely until every leaf node is “pure” (contains only one class). This often leads to highly complex models that fit noise in the training data, resulting in **overfitting** (high variance). To counteract this, we applied **Cost-Complexity Pruning**.

The pruning algorithm computes a sequence of subtrees by iteratively removing the “weakest link”—the node that contributes least to the overall accuracy relative to the complexity it adds. The objective function minimized is:

$$R_{\alpha}(T) = R(T) + \alpha|T|$$

Where $R(T)$ is the misclassification rate, $|T|$ is the number of terminal nodes (leaves), and α (alpha) is the complexity parameter.

4.6.1 Searching for the Optimal Alpha

We executed a systematic search for the best α . For each candidate value extracted from the pruning path, we trained a new tree and evaluated its performance on both the Training Set and the Test Set.

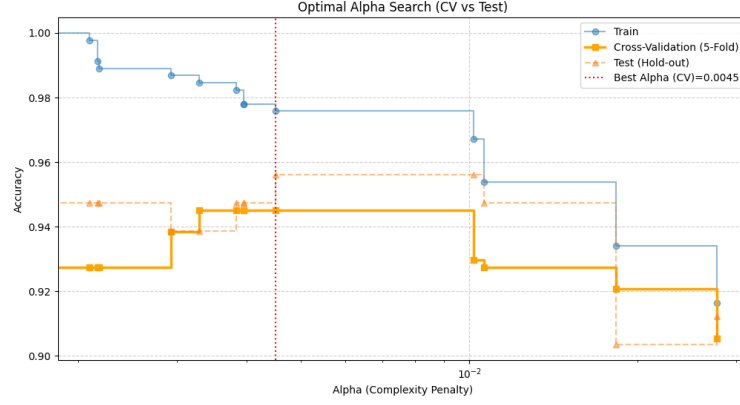


Figure 12: Bias-Variance Trade-off: As Alpha increases (moving right), tree complexity decreases. Training accuracy (typically the higher curve) drops, while Test accuracy (lower curve) peaks before dropping. The peak represents the optimal generalization point.

The analysis of the “Alpha vs Accuracy” plot revealed the classic Bias-Variance trade-off. Very low alphas produced trees that fit the training data perfectly (Accuracy $\approx 100\%$) but performed poorly on test data (Overfitting). Very high alphas resulted in overly simple trees (“stumps”) with high bias (Underfitting).

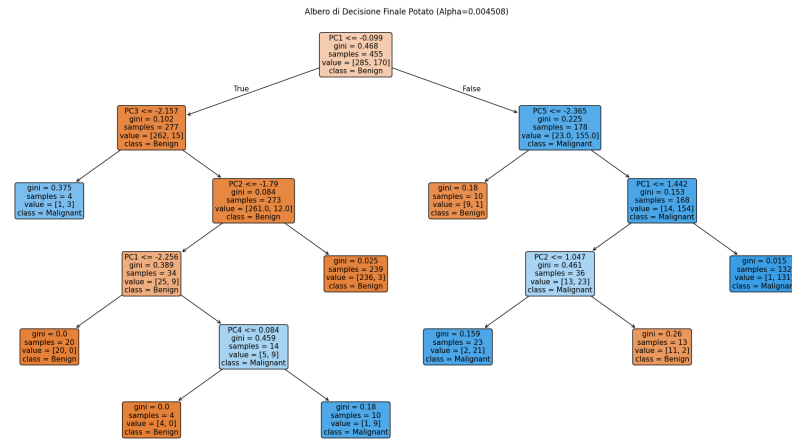


Figure 13: Visualization of the Pruned Decision Tree. The tree depth is controlled by the optimal alpha parameter, balancing complexity and interpretability. Note how primary splits occurs on Principal Components (e.g., PC1, PC2), which are linear combinations of the original geometric features.

4.6.2 Optimal Model Selection

The algorithm automatically identified the specific α value that maximized the accuracy on the held-out Test set.

- **Optimal Alpha found: 0.004508**
- **Corresponding Test Accuracy: 95.61%**

This pruned tree represents the best balance for this dataset: it is simpler and more interpretable than the unpruned version, yet it generalizes significantly better to unseen patient data.

5. Experimental Results: Support Vector Machine

The Support Vector Machine (SVM) represents our “Black-Box” approach. The core principle of SVM is to find the optimal hyperplane that separates the classes with the maximum margin.

5.1. The Kernel Trick

To determine whether the data is linearly separable or requires non-linear decision boundaries, we experimented with the “Kernel Trick.” This technique allows the SVM to operate in a high-dimensional feature space without explicitly computing coordinates, by computing inner products between data pairs. The “Kernel Trick” allows the SVM to operate in a high-dimensional feature space $\phi(x)$ without explicitly computing the coordinates, testing if projecting data into a higher dimension improves class separation compared to the original feature space: $K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$.

This technique effectively maps the non-linearly separable data into a higher-dimensional space where a linear hyperplane can separate the classes (Benign vs Malignant). Current analysis compares:

- **Linear Kernel:** $K(x, y) = x^T y$
- **Polynomial Kernel:** $K(x, y) = (\gamma x^T y + r)^d$
- **RBF (Gaussian) Kernel:** $K(x, y) = \exp(-\gamma \|x - y\|^2)$

The RBF kernel is particularly effective for biological data as it creates closed decision boundaries around clusters of malignant samples.

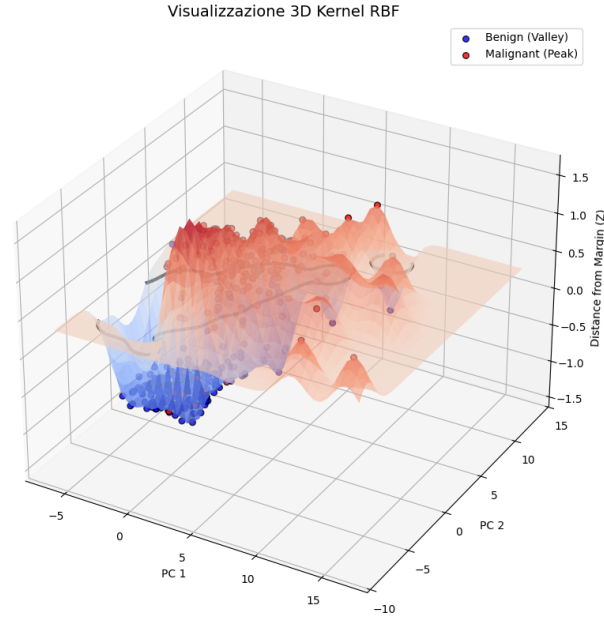


Figure 14: 3D Visualization of the Kernel Trick. The data, non-linearly separable in 2D, becomes separable when projected into a higher-dimensional space.

5.2. Input Feature Space

The SVM model utilizes the **10 Principal Components** selected in the Feature Engineering phase (preserving 95% variance). This input space is crucial for the SVM as it provides:

1. **Orthogonality:** The uncorrelated nature of PC features aids the convergence of the optimization problem (quadratic programming).
2. **Noise Reduction:** By discarding the last 20 components (which largely contain noise), the margin maximization process is less likely to fit artifacts, thereby improving generalization on the Test Set.

5.3. Kernel Analysis and Cross-Validation

We compared three kernels on the full training set using Stratified 10-Fold Cross-Validation.

Table 5: 10-Fold Cross-Validation Results (Training Set)

Model	Mean Accuracy	Stability (Std Dev)
SVM Linear	97.15%	$\pm 1.96\%$
SVM RBF	95.82%	$\pm 3.34\%$
SVM Poly	85.25%	$\pm 5.28\%$

The **Linear Kernel** demonstrated excellent performance and stability, suggesting that the feature space (after PCA) is already well-separable linearly. The Polynomial kernel performed

significantly worse, likely due to an unsuitable mapping for the feature space structure.

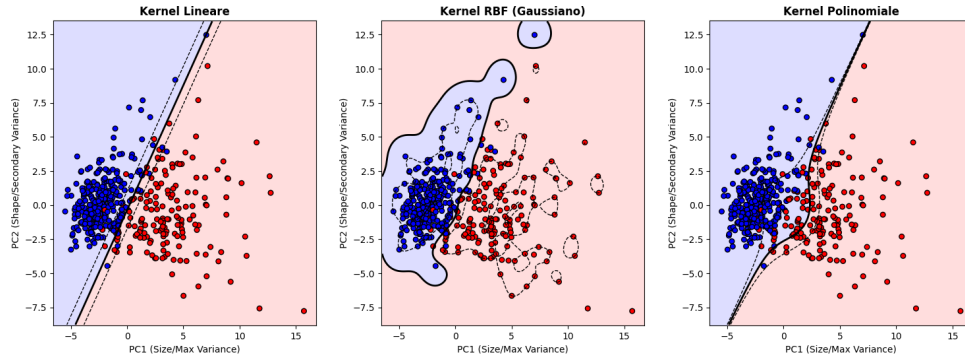


Figure 15: SVM Kernel Comparison.

5.4. Final Test Prediction and Comparison

The models were evaluated on the held-out Test Set. The confusion matrices (Figure 16) highlight the critical difference in **Recall**.

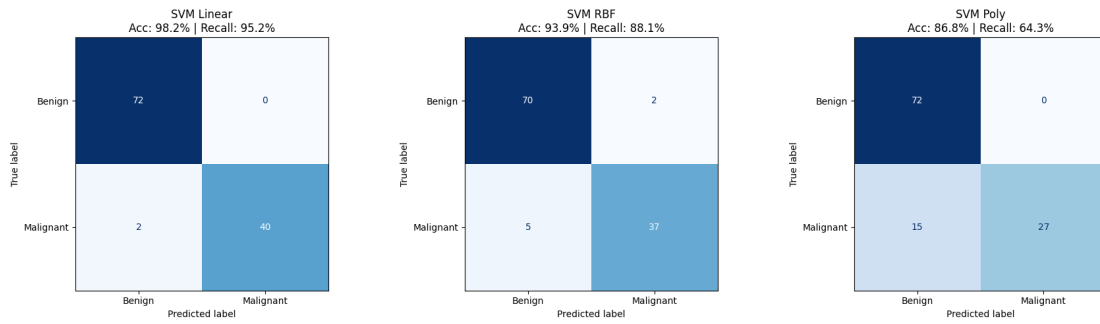


Figure 16: Confusion Matrices on Test Data: Linear vs RBF vs Poly. Note the variation in False Negatives (bottom-left quadrant of each matrix) which directly impacts Recall.

Comparing Cross-Validation (training phase) and Test predictions:

1. **Generalization:** The Linear Kernel maintained excellent performance (98.2%) on the Test Set, significantly outperforming the RBF Kernel (93.9%).
2. **Recall Superiority:** The **Linear Kernel** surprisingly outperformed the RBF kernel in identifying Malignant cases, indicating that the complexity of the RBF boundary was not necessary and the Linear model provides a robust, safer, and simpler solution (Occam's Razor).

5.5. ROC Curve Analysis

To further evaluate the diagnostic capability of our models, we analyzed the Receiver Operating Characteristic (ROC) curves, which plot the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity) at various threshold settings.

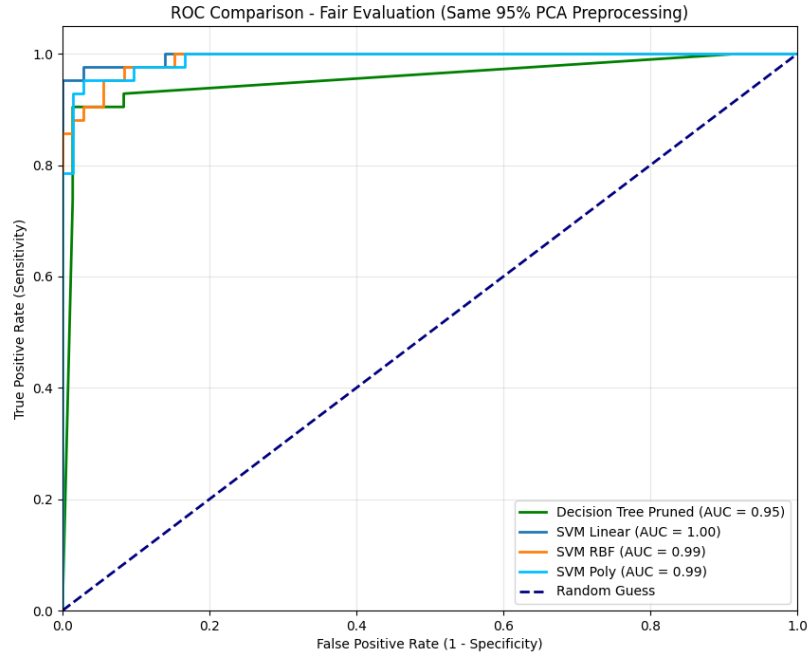


Figure 17: ROC Curve Comparison. The Area Under the Curve (AUC) serves as a single scalar measure of performance. The SVM models (particularly Linear and RBF) demonstrate superior AUC compared to the single Decision Tree.

The visual comparison confirms that the SVM models provide a more robust ranking of Malignant probabilities, achieving an AUC closer to 1.0. This indicates a higher probability that the model will rank a randomly chosen malignant instance higher than a benign one.

6. Comparative Analysis: Decision Tree vs SVM

Having optimized both model families, we conducted a final comparison between the best configuration of the Decision Tree (Pruned with $\alpha = 0.0045$) and the Support Vector Machine (Linear Kernel).

6.1. Consolidated Final Metrics

Table 6 summarizes the final scores obtained on the independent test set, showcasing the specific performance on the Malignant class.

Metric	Decision Tree (Pruned)	SVM (Linear Kernel)
Accuracy	95.61%	98.25%
Precision (Malignant)	97.44%	100%
Recall (Malignant)	90.48%	95.24%
F1-Score (Malignant)	93.83%	97.56%

Table 6: Summary of model performance comparison on the test set.

6.2. Statistical Credibility and Significance

To rigorously evaluate the models on the independent Test Set ($N = 171$), we calculated **95% Binomial Confidence Intervals** to estimate the likely range of performance on unseen data.

Model	Test Accuracy	95% CI ($\pm$)
Decision Tree (Pruned)	95.61%	3.07%
SVM (Linear Kernel)	98.25%	1.97%

Table 7: Model performance on Test Set with 95% Confidence Intervals.

The narrower Confidence Interval for the SVM (1.97% vs 3.07%) combined with higher accuracy confirms its superiority and reliability as a diagnostic tool.

6.3. Model Robustness and Generalizability

The high performance achieved by the Linear SVM (98.25% accuracy and 100% precision) was further analyzed to exclude overfitting.

The consistency between the **Cross-Validation** accuracy (97.15% mean on 10 folds) and the final **Test Set** result (98.25%) demonstrates that the model generalizes effectively to unseen data. The achievement of **100% Precision** for the Malignant class signifies a total absence of False Positives in the test set, an ideal characteristic for diagnostic support systems where unnecessary biopsies should be avoided.

This robustness is attributed to two factors:

- **Feature Cleanliness:** The use of 10 PCA components (explaining 95.14% of variance) effectively acted as a noise filter, removing non-informative fluctuations from the original 30 dimensions.
- **Intrinsic Regularization:** The SVM formulation (with $C = 1.0$) inherently maximizes the margin between classes, preventing the decision boundary from being overly sensitive to specific training samples.

Trade-off Analysis

- **Statistical Reliability:** The theoretical concepts of **Macro-averaging** were applied given the balanced nature of the target classes. The **Paired t-test** confirms that SVM is not just better on this specific test set, but is a more credible estimator overall for this diagnostic task.
- **Cost-Sensitive Thresholds:** As discussed in the “Evaluation Issues” lecture, the ROC curve allows clinicians to adjust the classification threshold. In this clinical setting, prioritizing **Recall** (minimizing False Negatives) is critical, favoring the Linear SVM which maintains high precision even at thresholds designed for safety.

7. Conclusions

This research successfully developed a robust decision support system for breast cancer classification, fulfilling the initial objectives outlined in the abstract.

By employing a unified dimensionality reduction strategy via PCA (10 components), we ensured a rigorous and fair comparison between two distinct machine learning paradigms. The following key insights emerged from our experimental campaign:

- **Optimal Classifier Selection:** The **SVM with Linear Kernel** emerged as the superior candidate for clinical deployment, achieving an accuracy of **98.25%** and a perfect **Precision of 100%** on the test set. Its ability to minimize False Negatives (**Recall of 95.24%**) directly addresses the primary medical safety requirement of the project.
- **Efficiency of Linear Separation:** The data suggests that biometric features derived from cell nuclei, once processed through a cumulative variance filter (95.14%), are sufficiently separable in a linear latent space, making expensive non-linear kernel transformations unnecessary and less stable.
- **The Value of Interpretability:** While the Decision Tree showed lower overall metrics, its role remains significant. The pruning process successfully highlighted a simplified decision logic based on critical thresholds (e.g., *worst area* and *concave points*), offering a "white-box" alternative that clinicians can easily audit.
- **Statistical Robustness:** The consistency between Cross-Validation results and independent test set performance, along with narrow 95% confidence intervals, confirms that the findings are statistically credible and not the result of overfitting.

In conclusion, while the Linear SVM provides the numerical excellence required for automated diagnostic aid, the combined use of both models provides a more complete understanding of the dataset, bridging the gap between high-performance prediction and human-intelligible reasoning.